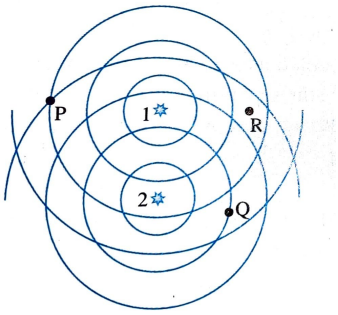
1. The figure shows the circular wavefronts emitted by two sources.



A. Are these sources in phase or out of phase?

B. Make a table with rows labeled P, Q, and R, giving the distances as multiples of λ and indicating, with a C or a D, whether the interference at the point is constructive or destructive.

2. Light $\lambda = 650$ nm passes through a double slit apparatus, with slits separated by 5 μm. What is the angle from the center of the device of the m=2 bright spot? What is the y position of the m=1.5 dark spot?

3. A double slit experiment is performed with light of wavelength 630 nm. The bright interference maxima are spaced by 1.8 mm on the viewing screen. What will the fringe spacing be if the light is cahnged to 420 nm.

4. Light from a sodium lamp ($\lambda = 589$ nm) illuminates two narrow slits. The spacing between consecutive maxima on a screen 150 cm behind the slits is 4.0 mm. What is the spacing in mm between the two slits?

5. The intensity pattern $I(y) = I_0 \cos^2(\alpha y)$, where $\alpha = \frac{\pi d}{\lambda L}$ Construct an intensity pattern where the spacing between maxima is about 2 mm. You will need to find values of λ, d, and L to make this so. Graph the resulting intensity pattern in geogebra.

6. Mapping Interference Patterns. Open `interference.html` in a web browser. This simulation shows circular wavefronts from two point sources. Click anywhere to measure the path lengths d_1 and d_2 from each source (in wavelengths) and the path difference $\Delta x = |d_1 - d_2|$.

A. Set the source separation to 4λ using the slider. Starting at the midpoint between the sources, click along a horizontal line moving to the right. Find and record 3 points where $\Delta x = 0$. What do you notice about these points?

B. Now find points where $\Delta x = 1\lambda$ (constructive, $m = 1$) and $\Delta x = 0.5\lambda$ (destructive). How are these points arranged spatially?

C. Turn on "Show Nodal Lines" and verify that your measured points lie on the green (anti-nodal) and red (nodal) lines.

D. Predict: if the source separation is $d = 6\lambda$, what is the maximum possible value of Δx? How many anti-nodal lines (including $m = 0$) will there be? Test your prediction by adjusting the slider.

7. Double-Slit Geometry. Open `double_slit.html` and select the "1. Geometry" tab. Use $d = 8$ μm, $\lambda = 500$ nm, and $L = 20$ cm.

A. Calculate λ/d in radians. This is the angular spacing between adjacent maxima for small angles.

B. Click on the intensity pattern to select the $m = 1$ bright fringe (the first bright band above center). The sidebar shows θ and $y = L\tan\theta$. Record both values. Verify that $d\sin\theta \approx 1\lambda$.

C. Turn on "Ripples" and "Nodal/Anti-nodal" lines. Notice how the green lines pass through crest-crest intersections (solid circles meeting solid circles), while the red lines pass through crest-trough intersections (solid meeting dashed). Click on a red nodal line and verify $d\sin\theta \approx 0.5\lambda$ (the $m = 0$ dark fringe).

D. Turn on "$\Delta = d\sin\theta$" to see the path difference triangle. Click on the $m = 2$ fringe. Sketch the triangle and label: the slit separation d, the path difference δ, and the angle θ.

E. Use the "Y-scale exaggeration" slider to go from 10× down to 1× (true scale). What happens to the apparent angle of the bright fringes? Why is exaggeration necessary for visualization when $\lambda \ll d$?

F. How can we even get the beams interfering at the point if they are effectively parallel? *Hint: Draw the wavefronts of each of the beams as plane waves. What is really interfering – is it the rays of light? Or something else?*